

Retrenchment or Reinvention?

Firm-Level Capability Reconfiguration under Trade Shocks

Giovanni Maran

Joint Doctoral Programme, University of Trento & Free University of Bozen-Bolzano

`giovanni.maran@unitn.it`

Supervisors: Chiara Tomasi & Stefano Schiavo (University of Trento)

The project in one paragraph. When a trade shock degrades the value of a firm's existing activities, does the firm *retrench* toward a coherent capability core, or *reinvent* itself by exploring new capability bundles? I measure the firm's export basket as a position in a firm-product *capability network*, split diversification into an *in-block* margin d_{in} (deepening the core) and an *out-of-block* margin d_{out} (exploring beyond it), and ask how a predetermined, continuous tariff shock moves the two. The static capability map is already built on Chinese customs microdata (§2); the causal panel is the next step (§3). I would value your view on the mechanism, on the identification, and on whether a tractable model can rationalise the sign (§4).

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1 The question and why it is open

The empirical trade literature has measured, in great detail, how tariff shocks move firm *performance*: productivity, markups, employment, survival (Pavcnik 2002; Topalova 2010; Amiti–Konings 2007; Le & Tomasi 2023). It is almost silent on what those shocks do to the *composition* of the firm's activity — which products are kept, dropped, or newly entered, and whether the surviving basket is a *deepening* of a related core or a *scattering* into new domains. This is the object I want to make causal.

The reason the sign is genuinely open is theoretical. March's (1991) exploitation–exploration tension says a firm whose core is threatened can either intensify what it knows (retrench) or search outward (explore), and his own result is that organisations systematically *over-exploit*. Several literatures sharpen this into a directional prediction:

- **Retrenchment (H1, the modal prior).** Sunk-cost and re-entry-cost hysteresis (Crozet et al. 2021), variance-minimisation through within-cluster deepening (Bottazzi–Secchi 2006; Vannoorenberghe 2016; Crozet et al. 2025), and the competition-to-coherence evidence of Valvano–Vannoni (2003) and the product-vector consolidation of Fontagné et al. (2018) all point to $d_{in} \uparrow$, $d_{out} \downarrow$, $Share_core \uparrow$.
- **Reinvention (H1', the frontier-firm tail).** Penrose (1959), Teece (1982), the moderate-shock search models of Matsusaka (2001) and Bernardo–Chowdhry (2002), and escape-competition (Aghion et al. 2005; Akcigit–Melitz 2022) leave open $d_{out} \uparrow$ for firms with slack and a defensible core.

The economic-complexity toolkit (Hidalgo–Hausmann 2009; Tacchella et al. 2012; Pugliese et al. 2019; Bruno et al. 2023; De Stefano et al. 2025) gives me the *measurement instrument* for capability structure — a block-modular firm–product partition — but it has been used almost exclusively descriptively. The strategic-dependency literature (Farrell–Newman 2019; Arjona et al. 2023; Korniyenko et al. 2017) identifies *which* products carry geopolitical leverage but has no firm-level causal test. The contribution I am aiming for sits in the empty intersection: *a causal estimate of the direction of capability adjustment to a predetermined trade shock, with the strategic-dependence content of the product as the key moderator.*

2 Preliminary evidence: the capability map of Chinese exporters

Before any causal claim, the measurement layer has to be real and economically legible. I build it on Chinese firm $\times$ product $\times$ destination $\times$ year customs exports. For the 2008–2015 window the firm–product–year panel has 11.9M rows over 372,030 firms and 4,657 HS6 products. To work with an established-exporter core I keep firms that export in *every* year of the window: 71,017 firms — only 19% of the count, but 62% of export value. The export basket is binarised with the firm-level Balassa index ($RCA_{fp} > 1$), giving a bipartite incidence with ~ 1.1 M links. Both firm diversification and product ubiquity are heavily right-skewed (Fig. 5 in the appendix), the usual complexity signature.

How the capability blocks are found (the algorithm, in plain terms). The firm–product matrix is a *bipartite* graph: firms on one side, products on the other, an edge whenever a firm has $RCA > 1$ in a product. A “capability block” is a set of firms *and* products that link among themselves far more densely than chance would predict. “Than chance” is made precise by a null model. Bipartite modularity Q_B (Barber 2007) scores a partition by the within-block edges *in excess* of those expected if every firm and every product kept its own number of links but rewired at random — the expected edge between firm f and product p is $k_f d_p / m$ (degree $\times$ degree over total links). BRIM is the heuristic that maximises Q_B . The *bipartite* null is the whole point: off-the-shelf optimisers (Louvain, Leiden) use a *one-mode* null that “expects” firm–firm and product–product edges which cannot exist in a bipartite graph; that mis-specified baseline inflates apparent within-block density and *over-segments* (here 13 and 9 blocks against BRIM’s 5). A resolution parameter γ scales the null “charge” for grouping two nodes: $\gamma = 1$ is standard modularity, $\gamma > 1$ charges more and returns more, smaller blocks. I use BRIM at $\gamma = 1$ as the baseline and report $\gamma = 2$ below.

Capability blocks are real, and they are not HS sectors. Maximising Barber’s (2007) bipartite modularity (BRIM) on the incidence matrix yields **five capability blocks** with a strong modular signal, $Q_B = 0.505$. The blocks have a clear economic identity (Table 1, Fig. 1): a large machinery and electrical *generalist* block (51% of firms, the assembly base), a paper/light-manufacturing block (23%), a near-pure *textiles* block (15%), a chemicals-plus-agri-food block

(11%), and a tiny ultra-specialised diamond satellite. The data-driven partition is *not* a re-labelling of HS sectors: against the HS-section partition it scores $\text{ARI} = 0.29$, $\text{NMI} = 0.43$, with blocks spanning on average thirteen HS sections. The block is a capability object, not an industry code — which is exactly what makes the $d_{\text{in}}/d_{\text{out}}$ split meaningful.

Table 1: The five capability blocks (2008–2015, HS6; 71,017 established exporters).

Block	Firms	Dominant content
B0	36,321 (51%)	Machinery & electrical 36%, base metals 24%, instruments 7% — the large <i>generalist</i> block
B2	16,095 (23%)	Paper 16%, misc. manufacturing, packaging — light manufacturing
B1	10,870 (15%)	Textiles 95% — apparel, cotton, man-made fibres (pure)
B3	7,691 (11%)	Chemicals 48% + agri-food $\sim 41\%$ (vegetables, foodstuffs, animals)
B4	40 (0%)	Precious 100% — diamonds (pure satellite)

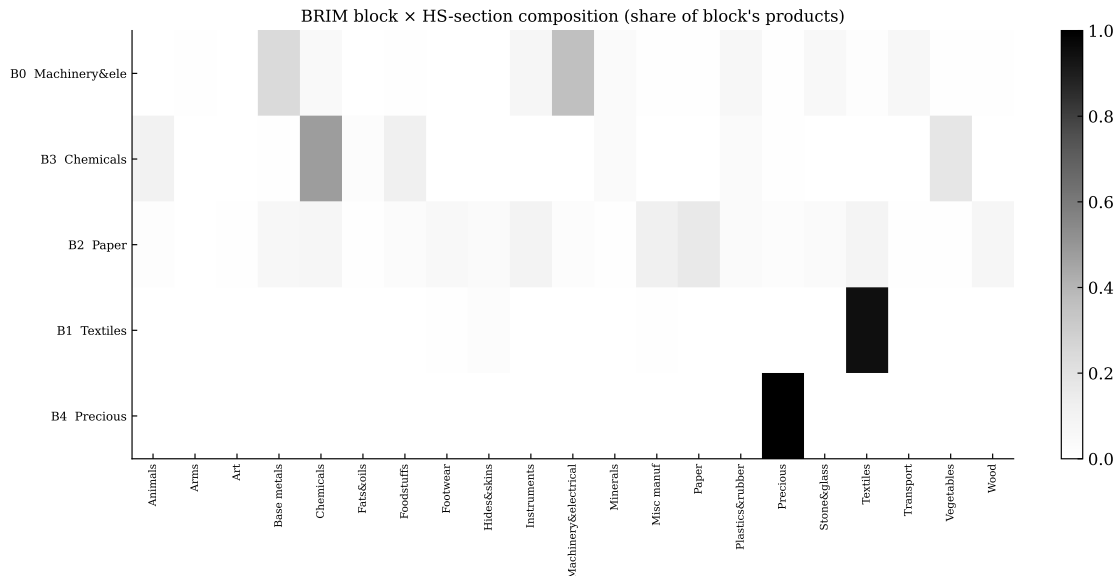


Figure 1: Capability block $\times$ HS-section composition. Textiles (B1) is near-pure; the generalist block (B0) spreads across machinery, metals and instruments. The off-diagonal mass is the evidence that blocks are not industry codes.

Does the resolution matter? ($\gamma = 1$ vs $\gamma = 2$). The block *count* is not an invariant — it depends on γ . To check that the substantive findings do not hinge on it, I re-ran the entire pipeline at the finer $\gamma = 2$, which penalises the null more and so returns more, smaller blocks (Table 2, Fig. 2). Some of the split is *economically meaningful*: agri-food, bundled with chemicals at $\gamma = 1$, separates into its own block, and dedicated footwear and transport blocks emerge. Some is *mechanical*: the pure textile block splits into two near-identical sub-blocks, while the machinery–metals generalist stays mixed at any resolution (it is where the bulk of firms sit). The point is that the *conclusions are invariant and only the levels rescale*: the structure is strongly modular and not a relabelling of HS sectors at both resolutions, and out-of-block exploration tracks destination reach rather than export value at both. What does move is the in/out accounting — finer blocks mean smaller in-block menus, so the out-of-block

share mechanically rises (pure-in-block 54.5% $\rightarrow$ 43.2%). The lesson I carry to the panel: *fix one partition choice and read d_{out} within a resolution, never across.*

Table 2: The same pipeline at two resolutions. Levels rescale; conclusions do not.

Quantity	$\gamma = 1$ (baseline)	$\gamma = 2$
BRIM blocks	5	10
Bipartite modularity Q_B	0.505	0.544
ARI vs HS sections	0.29	0.32
HS sections spanned per block (avg)	13	9.4
Pure in-block (Share_core = 1)	54.5%	43.2%
$\text{corr}(d_{\text{out}}, \text{dest. reach})$	0.18	0.16
$\text{corr}(d_{\text{out}}, \text{export value})$	0.03	0.06

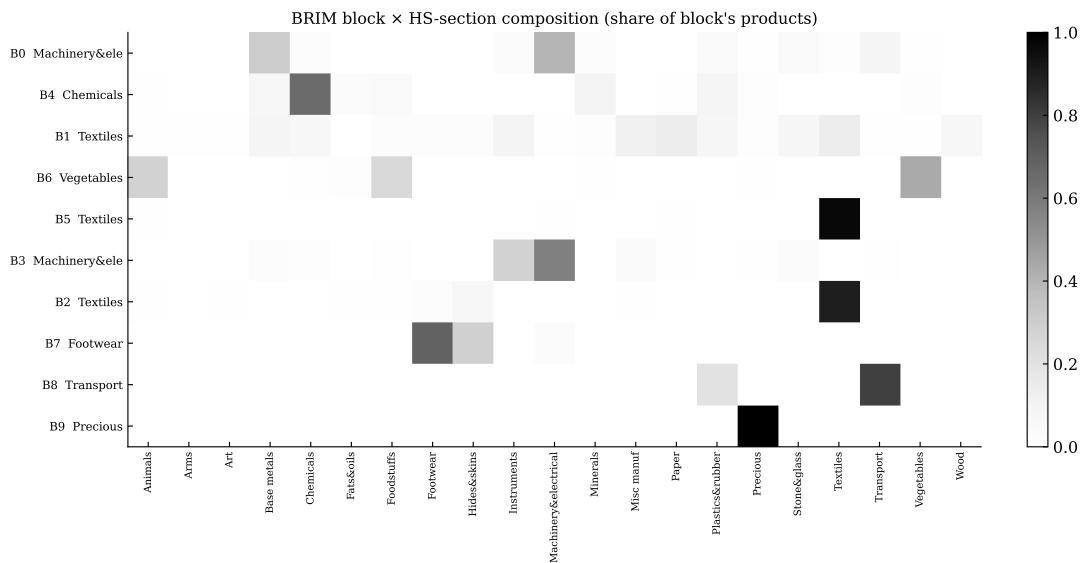


Figure 2: $\gamma = 2$: the ten finer blocks $\times$ HS sections (compare with the five-block $\gamma = 1$ map, Fig. 1). The newly isolated dark cells — a second textile block, a separated chemicals block, footwear, transport, precious — are the meaningful splits; the machinery generalist (and the light-manufacturing catch-all) stay spread, because they are genuinely entangled at any resolution.

These exporters are exploitation-heavy. Splitting diversification into d_{in} (competitive products *inside* the firm’s own block) and d_{out} (those *outside* it), the picture is one of deep specialisation (Table 3, Fig. 3): mean $d_{\text{in}} = 13.0$ against mean $d_{\text{out}} = 2.5$, and the out-of-block margin is *exactly zero for 54.5% of firms*. The concentration index $\text{Share_core} = d_{\text{in}} / (d_{\text{in}} + d_{\text{out}})$ has a median of 1.0 and a large mass at the ceiling. In March’s language, the established Chinese exporter grows along the d_{in} axis, within its capability block. **That distributional fact is what most shapes the panel design:** d_{out} is zero-inflated and floor/ceiling-bound, so the “reinvention” falsifier ($d_{\text{out}} \uparrow$) is better-powered than the “retrenchment” direction, and the outcome models have to respect the count/zero-mass structure (see §3).

Table 3: Diversification indicators ($N = 71,017$). $d_{\text{out}} = 0$ for the majority.

	mean	sd	p10	p25	median	p75	p90
d_{in}	13.01	20.83	2	4	7	14	29
d_{out}	2.53	11.38	0	0	0	2	5
Share_core	0.888	0.162	0.63	0.80	1.00	1.00	1.00

Pure in-block (Share_core = 1): 54.5%. Mixed: 45.5%. Pure out-of-block: 0%.

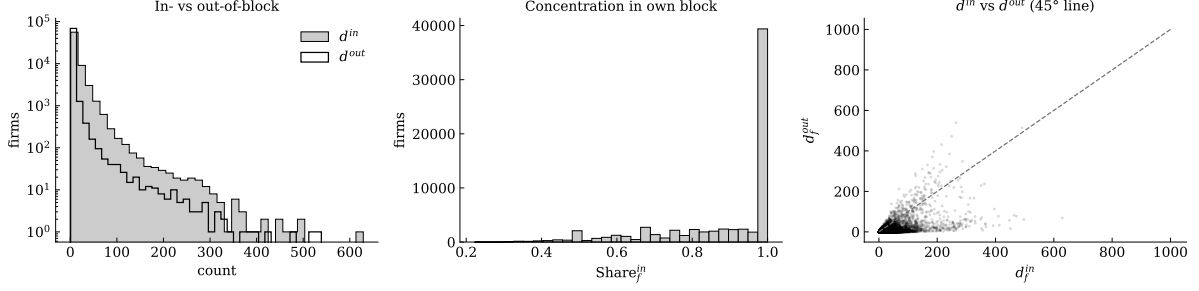


Figure 3: Left: d_{in} vs d_{out} (log y); d_{in} dominates, $d_{\text{out}} = 0$ for the median firm. Centre: Share_core, with a mass at 1 (pure specialists). Right: d_{in} vs d_{out} with the 45° line — the cloud lies below it, firms grow within their block.

A second pattern: out-of-block exploration is a *geographic* signature, not a size one. Because the microdata carry the destination, I can give d_{out} an economic identity. Across 71,017 firms, d_{out} correlates with *destination reach* (number of countries served) at 0.18, but with *export value* at only 0.03 (Spearman). Reaching more markets tracks out-of-block exploration; sheer revenue does not (Fig. 4). Holding destination reach fixed, scaling up in value actually *lowers* the out-of-block share (it deepens the core): a clean monotone pattern at the baseline resolution, though the one finding that partially reverses under a finer partition, so I read it cautiously. The message that survives: **the “explorer” firm is the wide-reach exporter, not the big-revenue one**, and exploration is concentrated in particular blocks (the paper/light-manufacturing block explores three times as much as any other). That is what tells me the eventual design must track the product and the geographic margins *jointly*, and must guard against export intermediaries, whose unrelated baskets produce spuriously high d_{out} and low portfolio coherence.

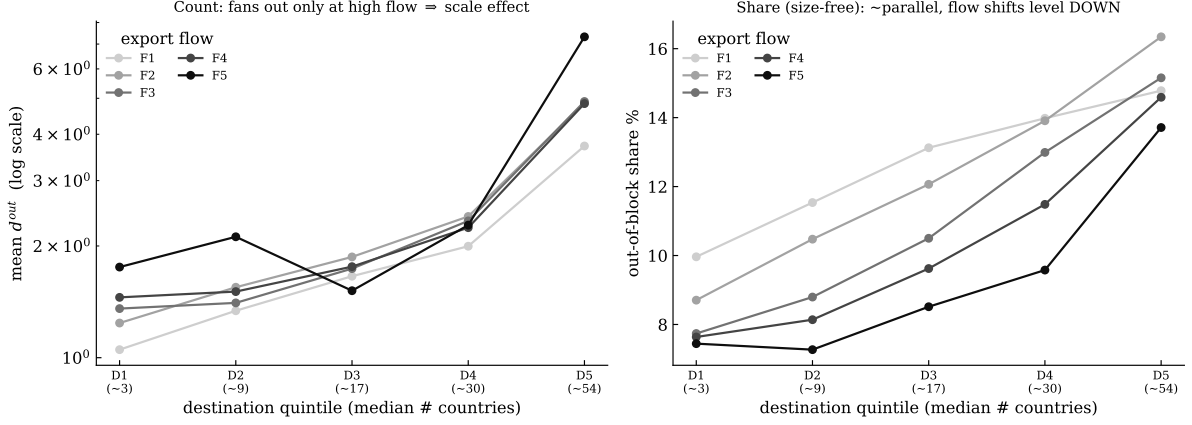


Figure 4: Out-of-block exploration by export-flow quintile (light = low to dark = high) $\times$ destination reach. **Left:** mean d_{out} count (log) — rises with destinations. **Right:** size-free out-of-block share — again rises with destinations, with the high-flow lines sitting *below*. Breadth tilts the portfolio outward; volume pulls it inward.

3 From description to causation: the problem, and where I need your intuition

The static layer is descriptive. The causal question (does an adverse trade shock push the firm toward d_{in} or d_{out} ?) needs a firm-by-year panel of the outcomes and a credibly exogenous shock. Two of the three moving parts I think are tractable; the third is genuinely open, and it is where I would most value your intuition. I lay out the problem and my candidate fixes, and then ask what *you* would do.

The shock — I think this part is in hand. I plan a continuous Bartik tariff dose

$$\text{TariffShock}_{it} = \sum_p \underbrace{s_{ip,pre}}_{\text{pre-shock shares}} \cdot \underbrace{\varepsilon_p}_{\text{Fontagné et al. 2022}} \cdot \underbrace{\Delta \ln(1 + \tau_{pt})}_{\text{Teti GTD 2024}},$$

estimated through a distributed-lag event study (leads = parallel-trends test, lags = hysteresis) with Adão et al. (2019) shift-share-robust standard errors. The exposure shares are pre-shock and the elasticities ε_p are external product parameters estimated on the world trade matrix, so they cannot be a function of the Chinese firms I study — this side of the design is predetermined by construction. I keep the dose *continuous* on purpose: discretising into treated/control throws away the exposure-intensity variation that identifies the dose-response (the cost is a stronger parallel-trends requirement, which I can probe but not fully test).

The hard part — turning a static map into a panel. Here is the crux, and the reason I want the algorithm understood (§2). The capability blocks are estimated on the *pooled* network. A panel needs the outcome to vary over time, but d_{in}/d_{out} are defined *relative to the partition*, so the instant the partition itself moves, the outcome and its own measuring instrument move together. There are two routes, each with a cost:

- **(a) Re-estimate the partition every year.** Conceptually attractive (capabilities really do evolve), but it makes block membership co-move with the shock: a tariff that pushes a firm off a product can show up as the *product* switching blocks, so the treatment is absorbed *into* the measuring instrument instead of being read *against* it. Worse, the temporal-coupled multilayer machinery (Mucha et al. 2010) one would use to smooth the yearly slices *couples adjacent years*, so the block at t depends on $t+1$: a look-ahead that disqualifies it as a clean

right-hand-side panel object. (It is also computationally heavy: BRIM on each yearly slice at the full $\sim 460k$ -firm scale.)

- **(b) Estimate the partition once on pre-shock years and hold it fixed**, letting only the firm’s RCA incidence vary. This keeps the measuring instrument exogenous to the shock, at the cost of a map that grows stale — classical measurement error, which attenuates *toward* zero. That is an honest bias, not a confound, but it is a real loss of power.

My current lean is (b) as the baseline, with a *pre-shock-evolving* map (estimated only on pre-shock slices) as robustness, and — the option I am most drawn to but least sure of — a binary, staggered design whose *not-yet-treated* firms could anchor a partition that evolves *contemporaneously yet stays exogenous*, in the spirit of a clean control group. **This single choice is the thing I would most like your view on:** is the cleaner estimand worth a held-fixed (and therefore stale) map, or is there a smarter way to let the capability geometry move *with* the data without letting it move *with the treatment*? Put concretely, it is a question about the *detection algorithm* itself: can BRIM (or any community-detection method) be made to deliver a partition that updates with the data yet stays, by construction, blind to the shock? Any intuition you have on that — even that I am framing it wrongly — would help.

What the preliminary data already dictate (whatever I decide above). The static layer is not just motivation; it pins down choices. (i) d_{in} is the cleanest outcome; d_{out} is 54.5% zeros $\Rightarrow$ a count/PPML model, not OLS. (ii) The ceiling at $Share_core = 1$ means the retrenchment direction is poorly powered and the $d_{out} \uparrow$ falsifier is the sharper test. (iii) Effects should vary by destination reach and by block, since that is where exploration lives. (iv) Coherence flags intermediaries (unrelated baskets, spurious d_{out}) and must be a control. (v) The balanced core is economically dominant but selected — external validity to the entry/exit tail is a stated caveat, and it interacts with the exit problem below.

4 Open issues, and the questions I would like your view on

I separate what I already see as the *soft spots* from the *questions* I would genuinely like to put to you. I have grouped the questions by mechanism, by empirical design, and by theory, but the most valuable feedback would be on which of these is the binding constraint on the project.

Soft spots I am already worried about

1. **The within-China RCA denominator under an aggregate shock.** Blocks are built from the Chinese firm–product network, so a China-relative Balassa is the internally consistent benchmark — but a sector-wide tariff shock moves the aggregate Chinese export of a treated product for *all* exposed firms, mechanically shifting threshold crossings. I treat this as an inference issue (leave-one-out and ex-China BACI denominators as robustness), but it is really a *general-equilibrium* contamination of my own measure.
2. **Exit selection.** The shock induces exit (Crozet et al. 2021); exited firms have no d_{in}/d_{out} , so survivors are selected. I plan an exit equation plus Lee/IPW bounds and read the headline as conservative — but the balanced-sample design already conditions on survival, which may understate the very margin I care about.
3. **Spillovers / SUTVA.** A tariff on one product reallocates demand to substitutes held by *other* firms in the same block; Adao et al. (2019) absorbs the correlation in the SE but not the causal interference.
4. **Zero-inflation and power.** With $d_{out} = 0$ for the majority, the retrenchment direction may be statistically hard to move; the design may end up better at *rejecting* reinvention than *confirming* retrenchment.

(A) Questions on the mechanism

1. Is retrenchment to the core best read as *defensive pruning* (shedding the cheapest-to-cut peripheral products) or *active consolidation* (reinvesting in the core)? These have the same `Share_core` signature but opposite welfare readings. Is there a margin in the data — e.g. the complexity of the products *dropped* (Nomaler–Verspagen 2023) — that can separate them?
2. The destination finding suggests d_{out} is partly a *geographic* object. Should the unit of analysis be the firm–product matrix (following the capability-network tradition) or a firm–product–destination tensor? Does promoting to the tensor buy identification, or just noise?
3. Is “exploration” through d_{out} genuinely capability-building, or is a large share of it intermediary churn that coherence controls cannot fully remove?

(B) Questions on the empirical design / how to innovate the analysis

1. **The held-fixed-vs-evolving map trade-off** (the open seam of §3). Beyond the two routes I sketched, is there a third I am missing — for instance a *partition-free* outcome (a continuous relatedness or coherence score of the basket) that would sidestep the timing problem altogether, at the cost of losing the crisp in/out split? Which would you reach for?
2. **Parallel trends for a continuous dose.** Beyond plotting lead coefficients, what is the most convincing way to defend “no selection into dose level” when the dose is a smooth shift-share and there is no clean control group?
3. **The denominator problem.** Is the within-China RCA benchmark defensible under an aggregate shock, or should I move to an ex-China (rest-of-world) benchmark as the headline and pay the cost of a less internally consistent block definition?
4. Given the zero-inflation, is there a better-powered *single* outcome (a product-level retention hazard, in the spirit of Fontagné et al. 2018) than the firm-level `Share_core`?

(C) A theoretical model — here I am more interested in what occurs to you

On the theory side I have instincts but no commitment, and this is where your reaction matters most — so these are deliberately open.

1. If you wanted to rationalise an in-block/out-of-block margin, *what backbone would you reach for?* My own instinct drifts toward a multi-product firm in the “core competence” tradition (Eckel–Neary 2010; Mayer–Melitz–Ottaviano 2014), where products sit on a ladder ranked by distance from the firm’s core competence and tougher competition skews the mix *toward* the core — which is exactly my retrenchment prediction, with in-block $\approx$ near-core and out-of-block $\approx$ far-from-core. The honest gap is that their ladder is ordered by *cost/productivity*, whereas my capability blocks are ordered by *relatedness* (network co-occurrence): is “distance from core” the same object in the two senses, or do I need a genuine capability/relatedness model instead? That is the piece I would most like your reaction to.
2. Is there a natural way to make the *sign* endogenous — retrenchment for most firms, reinvention for a high-capability tail — as a comparative static in a single state variable (capability stock, or distance-to-core)? That is the form in which March’s (1991) verbal tension would become testable, but I do not yet see the minimal model that gives it.
3. Does any of this *need* general equilibrium? The within-China RCA denominator is itself a GE object, so the aggregate-shock “contamination” of my measure (soft spot 1) might be better treated as a structural feature inside a quantitative trade model — which would also let me speak to the *welfare* of retrenchment, not just its sign. Is that worth the modelling cost, or a distraction from a clean reduced-form paper?

4. If you did build it, which empirical moment would you discipline it on first — the dose-response slope, the $d_{\text{in}}/d_{\text{out}}$ heterogeneity by strategic dependence, or the destination-reach gradient?

What I would most like to leave the meeting with. A judgement on the binding constraint: is this project limited by *identification* (the map-timing choice and the continuous dose), by *measurement* (the RCA denominator and intermediaries), or by the absence of a *theoretical* object that makes the $d_{\text{in}}/d_{\text{out}}$ split a structural quantity? My current bet is that the empirics can carry a first paper on their own, with the model as a second-stage companion — but I would like to pressure-test that ordering with you.

A Supporting figure: heavy-tailed structure

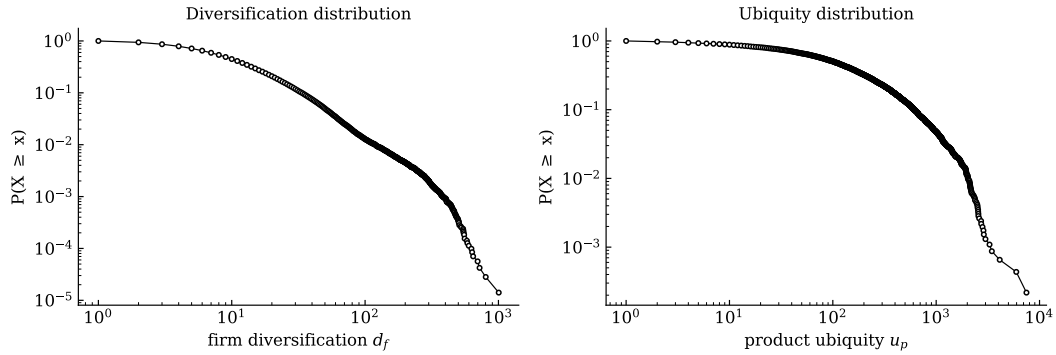


Figure 5: Complementary CDFs of firm diversification d_f (left) and product ubiquity u_p (right), log-log axes. Most firms are specialised and a few hugely diversified; most products niche and a few near-universal — the standard economic-complexity signature, which the block structure then organises.

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